
3.2.4 The variety of life is extensive and this is reflected in similarities and differences in its biochemical basis and cellular organisation.

Haemoglobin

The haemoglobins are a group of chemically similar molecules found in many different organisms.

Haemoglobin is a protein with a quaternary structure.

The role of haemoglobin in the transport of oxygen.

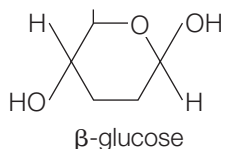
The loading, transport and unloading of oxygen in relation to the oxygen dissociation curve.

The effects of carbon dioxide concentration.

Candidates should be aware that different organisms possess different types of haemoglobin with different oxygen transporting properties. They should be able to relate these to the environment and way of life of the organism concerned.

Carbohydrates

The structure of β -glucose as



and the linking of β -glucose by glycosidic bonds formed by condensation to form cellulose.

The basic structure and functions of starch, glycogen and cellulose and the relationship of structure to function of these substances in animals and plants.

Cells

There are fundamental differences between plant cells and animal cells.

The structure of a palisade cell from a leaf as seen with an optical microscope.

The appearance, ultrastructure and function of

- cell wall
- chloroplasts.

Candidates should be able to apply their knowledge of these and other eukaryotic features in explaining adaptations of other plant cells.
